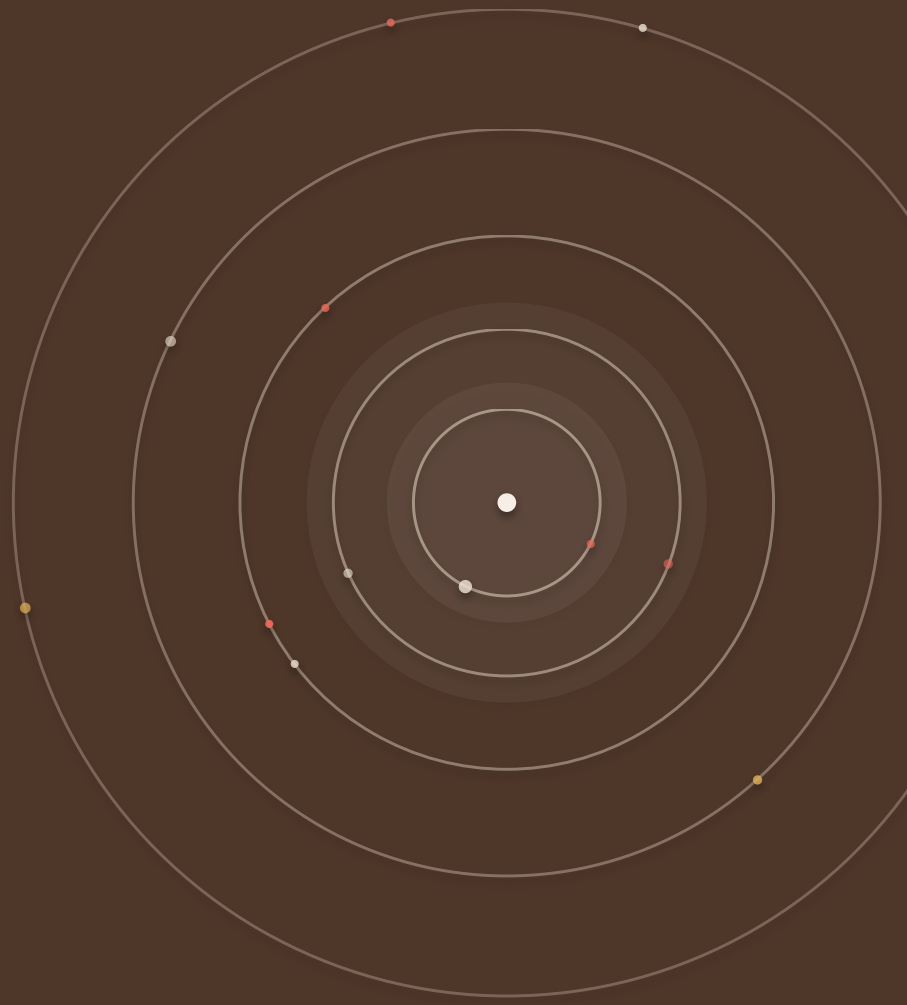
BROWN

CosmoAgent

An AI Agent for Cosmology



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Brown University · Brown Center for Theoretical Physics and Innovation

with Stephon Alexander, Loukas Gouskos, Savvas M. Koushiappas and Cooper Niu

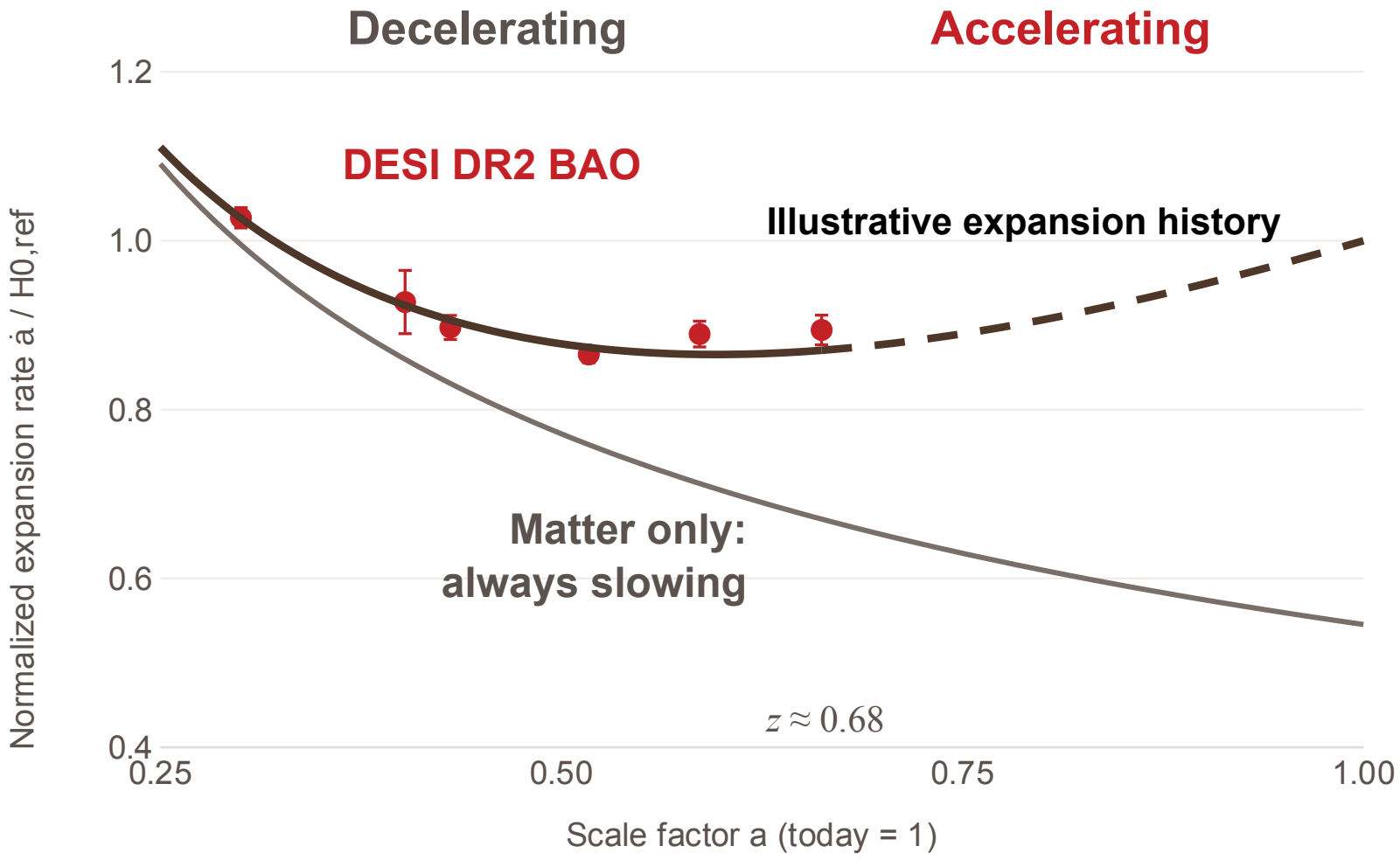
arXiv:2610.16536

Inspiration

autoresearch A. Karpathy, author of nanoGPT		**CosmoAgent** the same loop for physics
Change the small GPT training code	**Propose**	Propose a theory (an action)
Train the model for five minutes	**Test**	Derive the equations and solve them
Check the validation loss	**Score**	Fit to real supernova and BAO data
Loss goes down: keep. If not: undo.	**Decide**	Memory guides the next proposal

One difference: our agent only picks grammar-allowed, well-defined theories.

Late-time expansion



In general relativity

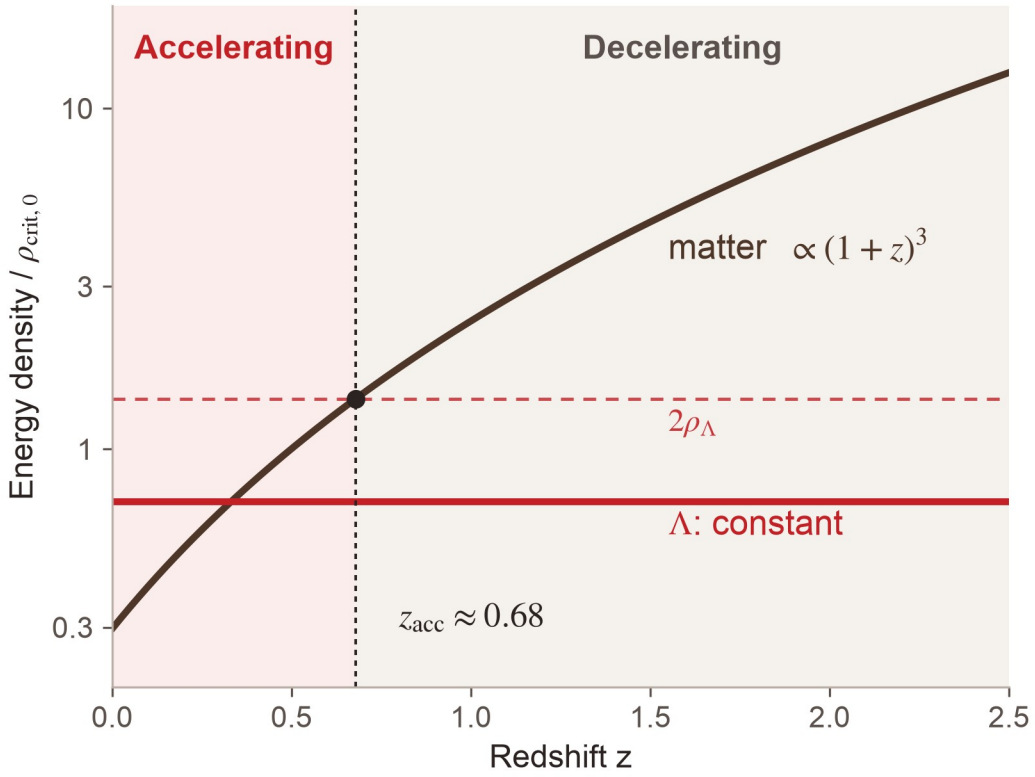
$$\ddot{a}/a = -(4\pi G/3)(\rho_{\rm tot} + 3p_{\rm tot})$$

Matter slows expansion.

Acceleration requires $\rho_{\rm tot} + 3p_{\rm tot} < 0$.

What drives cosmic acceleration?

Dark Energy and ΛCDM



1 **Friedmann equation (flat, late times)**

$$H^2 = (8\pi G/3)\,(\rho_m + \rho_\Lambda)$$

2 **Energy conservation: each part dilutes differently**

$$\dot{\rho} + 3H(\rho + p) = 0 \quad \Rightarrow \quad \rho \propto a^{-3(1+w)}$$

$$\rho_m \propto a^{-3} \quad (w = 0), \qquad \rho_\Lambda = \text{const} \quad (w = -1)$$

3 **In redshift, a = 1/(1+z): one shape parameter, Ωm**

$$H^2(z) = H_0^2\,[\,\Omega_m(1+z)^3 + 1 - \Omega_m\,]$$

4 **Acceleration (ä > 0) starts when ρm < 2ρΛ**

$$(1+z_{acc})^3 = 2(1-\Omega_m)/\Omega_m \quad \Rightarrow \quad z_{acc} \approx 0.68$$

Flat ΛCDM: one shape parameter, Ωm ≈ 0.3. This is our reference model.

Dark energy have dynamics?

The reference: ΛCDM

$$\mathcal{L}_{\Lambda\mathrm{CDM}} = \frac{M_p^2}{2} R - \rho_\Lambda + \mathcal{L}_m$$

$$w \equiv p/\rho \qquad w_\Lambda = -1$$

The question: dynamics?

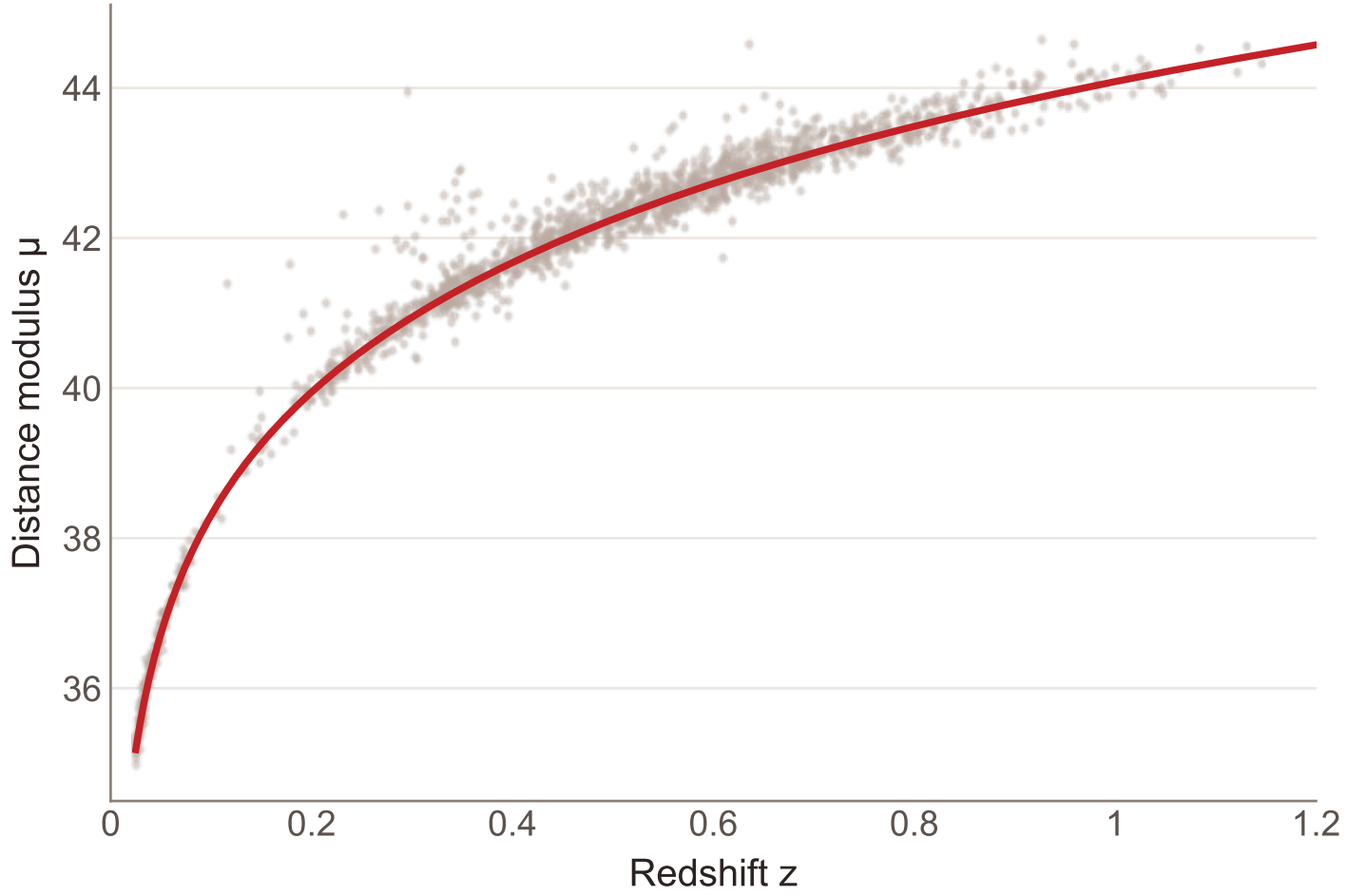
A rolling scalar field?

A changing gravitational coupling?

Higher-curvature interactions?

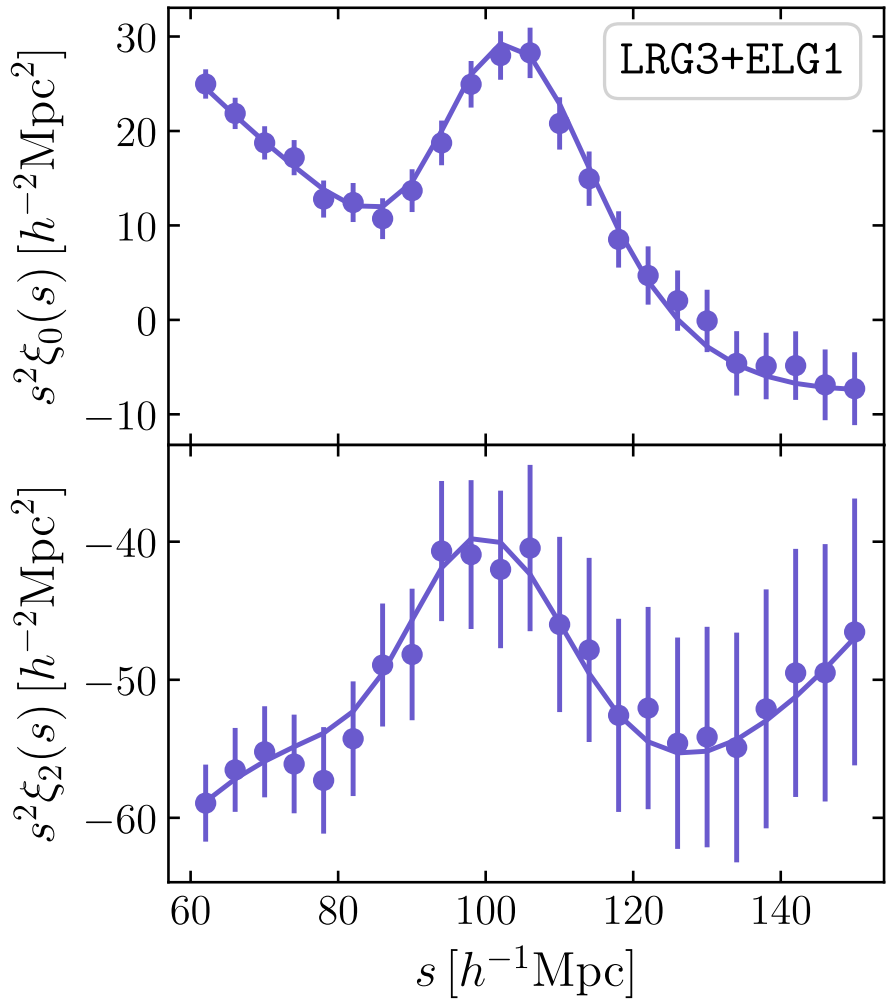
DES measures distance from supernova brightness

Dark Energy Survey



$$\mu = 5\log_{10}[d_L/\mathrm{Mpc}] + 25$$

DESI measures a standard ruler in galaxy clustering

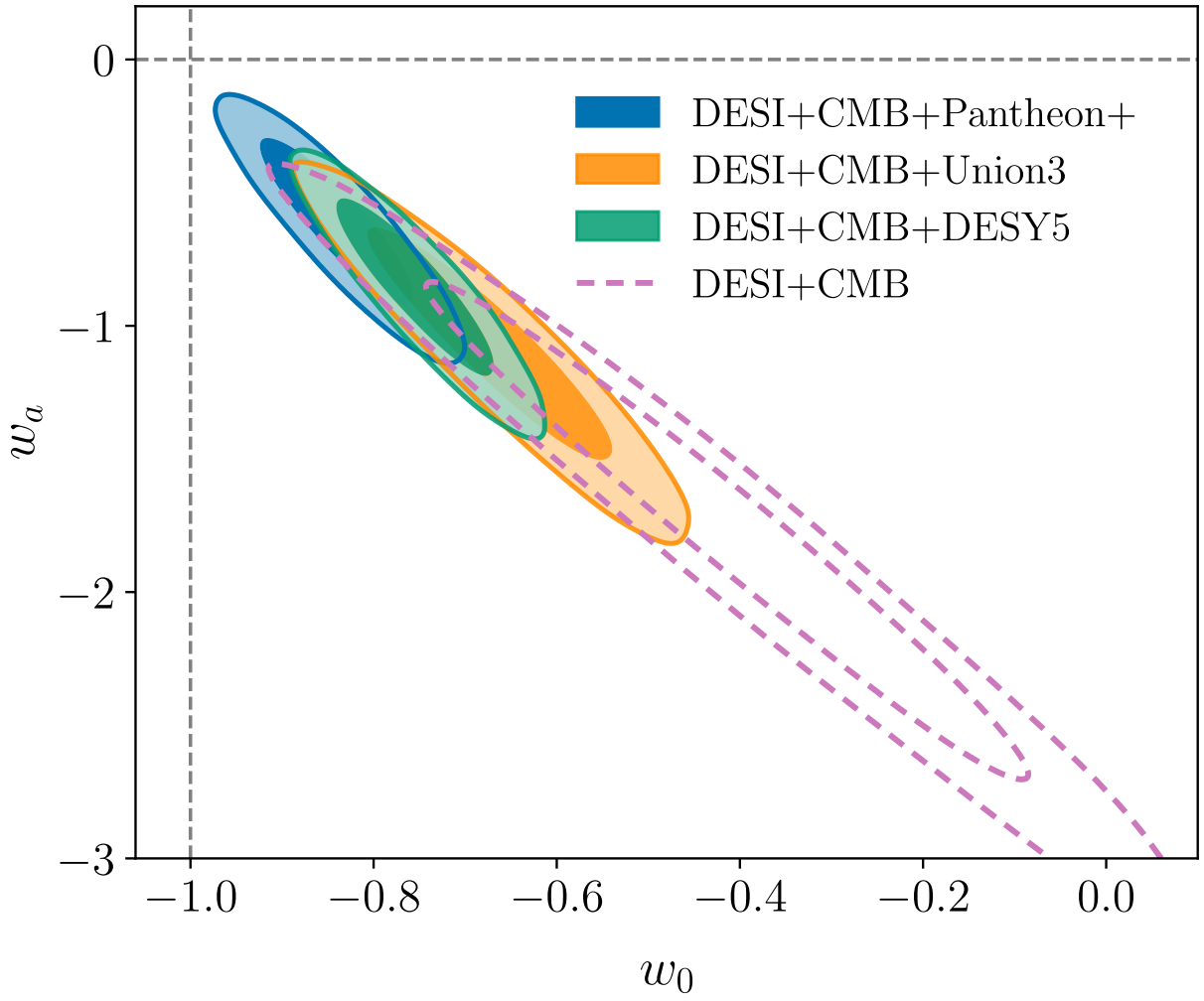


Dark Energy Spectroscopic Instrument

$$\theta_{\mathrm{BAO}} \simeq \frac{r_d}{D_M(z)}, \qquad \Delta z_{\mathrm{BAO}} \simeq \frac{r_d H(z)}{c}$$

DESI DR2 (2025), Fig. 5: LRG3 + ELG1, 0.8 < z < 1.1 • arXiv:2503.14738

DESI's dark energy hint motivates a search for dynamics



$$w(a) = w_0 + w_a(1-a)$$

CPL parameterization
w_0: equation of state today
w_a: evolution with scale factor

$$\Lambda\text{CDM}: \ (w_0, w_a) = (-1, 0)$$

Contours favor $w_0 > -1$, $w_a < 0$

Published DESI DR2 + CMB + SN posteriors (2025), Fig. 11. arXiv:2503.14738v2

Models for dark energy

Cosmological constant

$$\mathcal{L}_{\mathrm{DE}} = -\rho_{\Lambda}$$

Quintessence

$$\mathcal{L}_{\mathrm{DE}} = -\tfrac{1}{2}(\nabla\phi)^2 - V(\phi)$$

k-essence

$$\mathcal{L}_{\mathrm{DE}} = P(X,\phi)$$

Scalar–tensor / curvature

$$\tfrac{1}{2}F(\phi)R, \quad \xi(\phi)\mathcal{G}, \quad \ldots$$

A covariant grammar lets us explore many more combinations of these ingredients.

FLRW spacetime

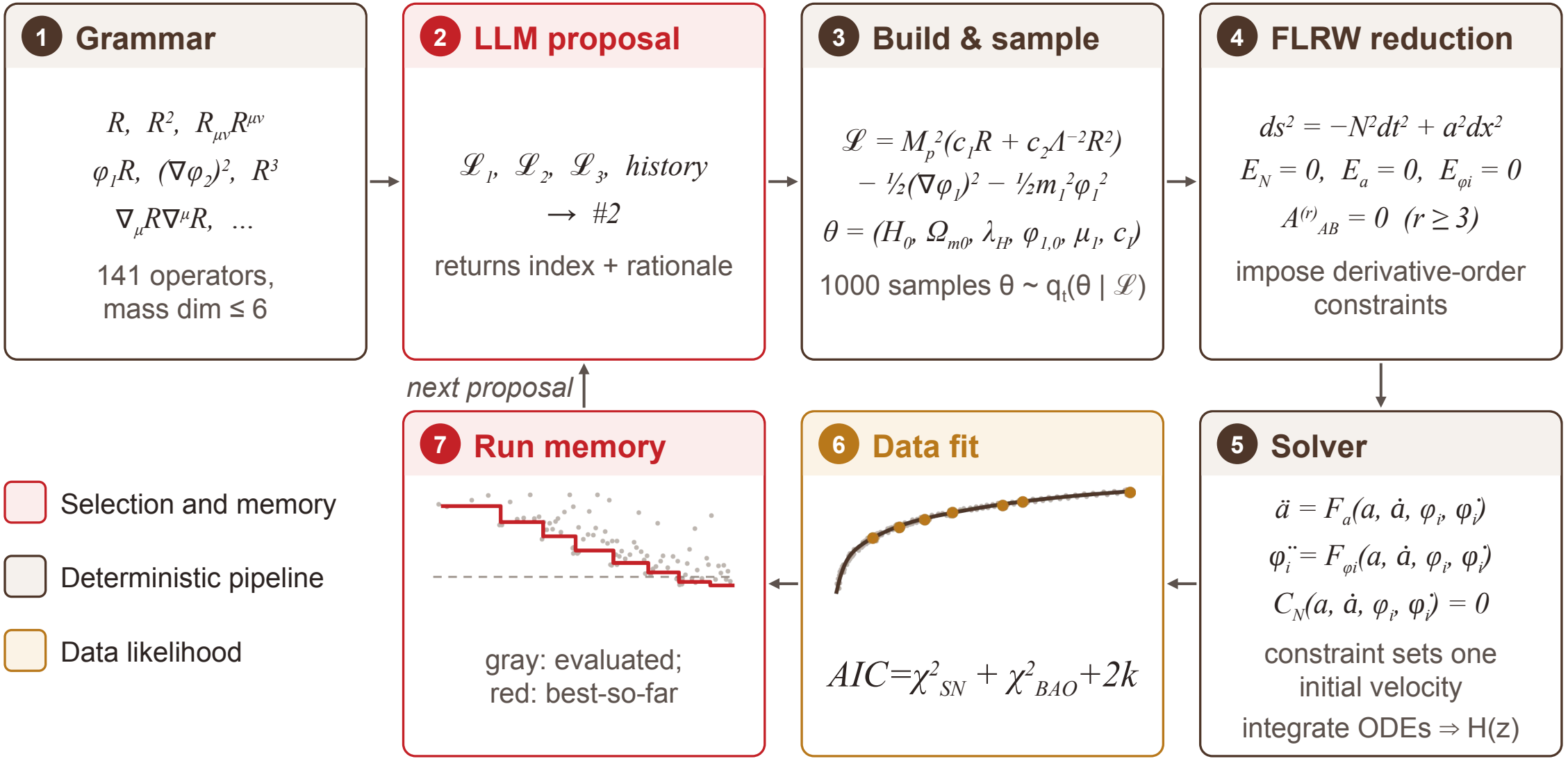
$$S \supset \int d^4x \sqrt{-g}\, [\ \textit{gravity \& dark energy}\ ]$$

This is the part we search

We study cosmic expansion on a spatially flat FLRW background:

$$ds^2 = g_{\mu\nu}\, dx^\mu\, dx^\nu = -dt^2 + a^2(t)\, d\mathbf{x}^2$$

From a proposed action to observational feedback



A covariant grammar defines the search space

$$\mathcal{L} = M_p^2 \sum_{I \in \mathcal{G} \cup \mathcal{N}} c_I \Lambda_{\text{cutoff}}^{2-d_I} \mathcal{O}_I + \sum_{I \in \mathcal{S}} c_I \Lambda_{\text{cutoff}}^{4-d_I} \mathcal{O}_I + \mathcal{L}_{\text{can}}$$

Pure curvature

$R,\ R^2,\ R_{\mu\nu}R^{\mu\nu},\ \ldots$

Scalar–curvature

$\phi R,\ \phi^2 R,\ (\Box\phi)R,\ \ldots$

Pure scalar

$\phi^2,\ \phi^4,\ (\nabla\phi)^2,\ \ldots$

The EFT cutoff supplies the powers needed to balance mass dimensions.

141 candidate operators, organized by mass dimension

$$[R] = 2, \qquad [\nabla] = 1, \qquad [\varphi] = 1$$

Dimension	Representative operators	Count
2	$R,\ \varphi_1{}^2 \ldots$	4
3	$\varphi_1 R,\ \varphi_1{}^3 \ldots$	8
4	$R^2,\ \varphi_1{}^2 R,\ (\nabla\varphi_1)^2 \ldots$	19
5	$\varphi_1 R^2,\ \varphi_1{}^5 \ldots$	36
6	$R^3,\ \varphi_1{}^2 R^2,\ \varphi_1{}^6 \ldots$	74

Each candidate action combines operators from this pool.

Two scalar fields

No scalar	φ_1 only	φ_2 only	Both fields

Canonical kinetic terms + a choice of potential

$$V_i = \frac{1}{2} m_i^2 \phi_i^2 \qquad \text{or} \qquad m_i^2 f_i^2 [1 - \cos(\phi_i / f_i)]$$

Field normalizations

$$\phi_1 = M_p \varphi_1, \qquad \phi_2 = \Lambda_{\text{cutoff}} \varphi_2$$

Structures and numerical parameters are explored together through the search loop.

Derive the background equations, then filter

1 **a homogeneous geometry in minisuperspace**

$$ds^2 = -N^2(t)\,dt^2 + a^2(t)\,d\mathbf{x}^2, \qquad \phi_i = \phi_i(t)$$

---

2 **Vary first; then choose cosmic time N = 1**

$$E_a = \frac{\delta S}{\delta a} = 0, \qquad E_{\phi_i} = \frac{\delta S}{\delta \phi_i} = 0, \qquad E_N = \frac{\delta S}{\delta N} = 0$$

---

3 **Cancel third and higher time derivatives （Ostrogradsky ghost）**

$$E_A \text{ contains no } y_B^{(r)} \text{ for } r \geq 3$$

Data likelihood

1,820

DES-Dovekie supernovae

13

DESI DR2 BAO measurements

Background EOM → H(z) → SN and BAO distances

$$\chi^2 = \chi^2_{\mathrm{SN}} + \chi^2_{\mathrm{BAO}}, \qquad \mathrm{AIC} = \chi^2_{\mathrm{min}} + 2k$$

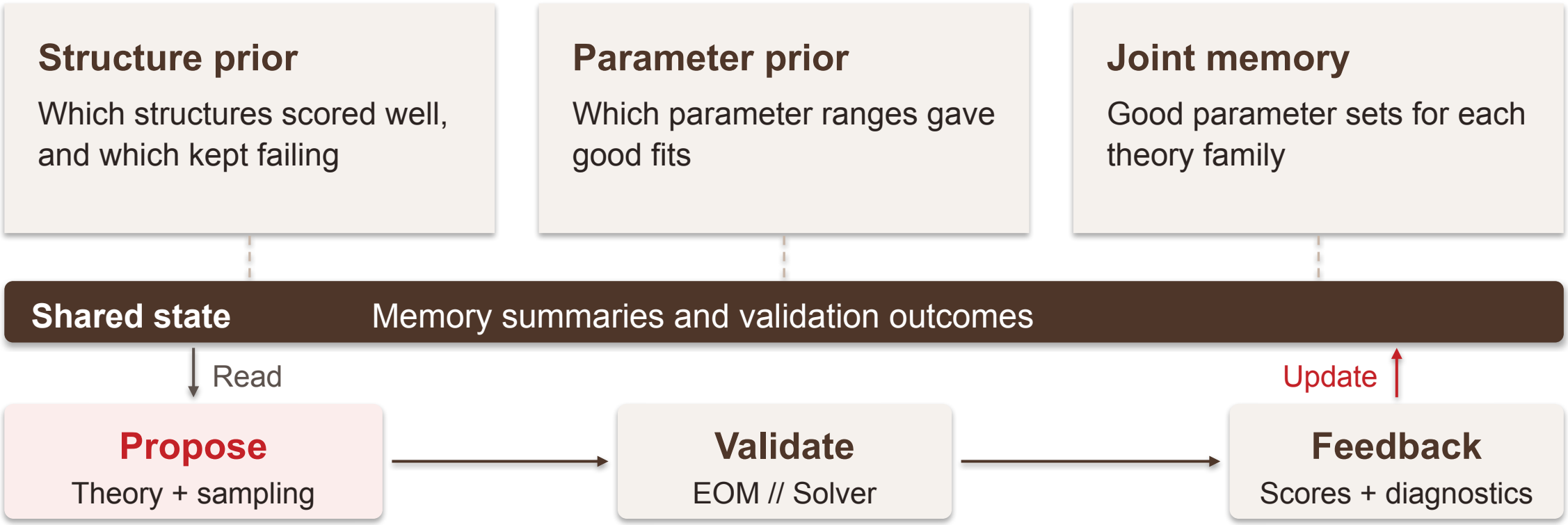
Profile the common SN offset and BAO scale.

Flat ΛCDM reference: $\chi^2 = 1645.28$ • $k = 3$ • AIC = 1651.28

The search adapts using current-run feedback

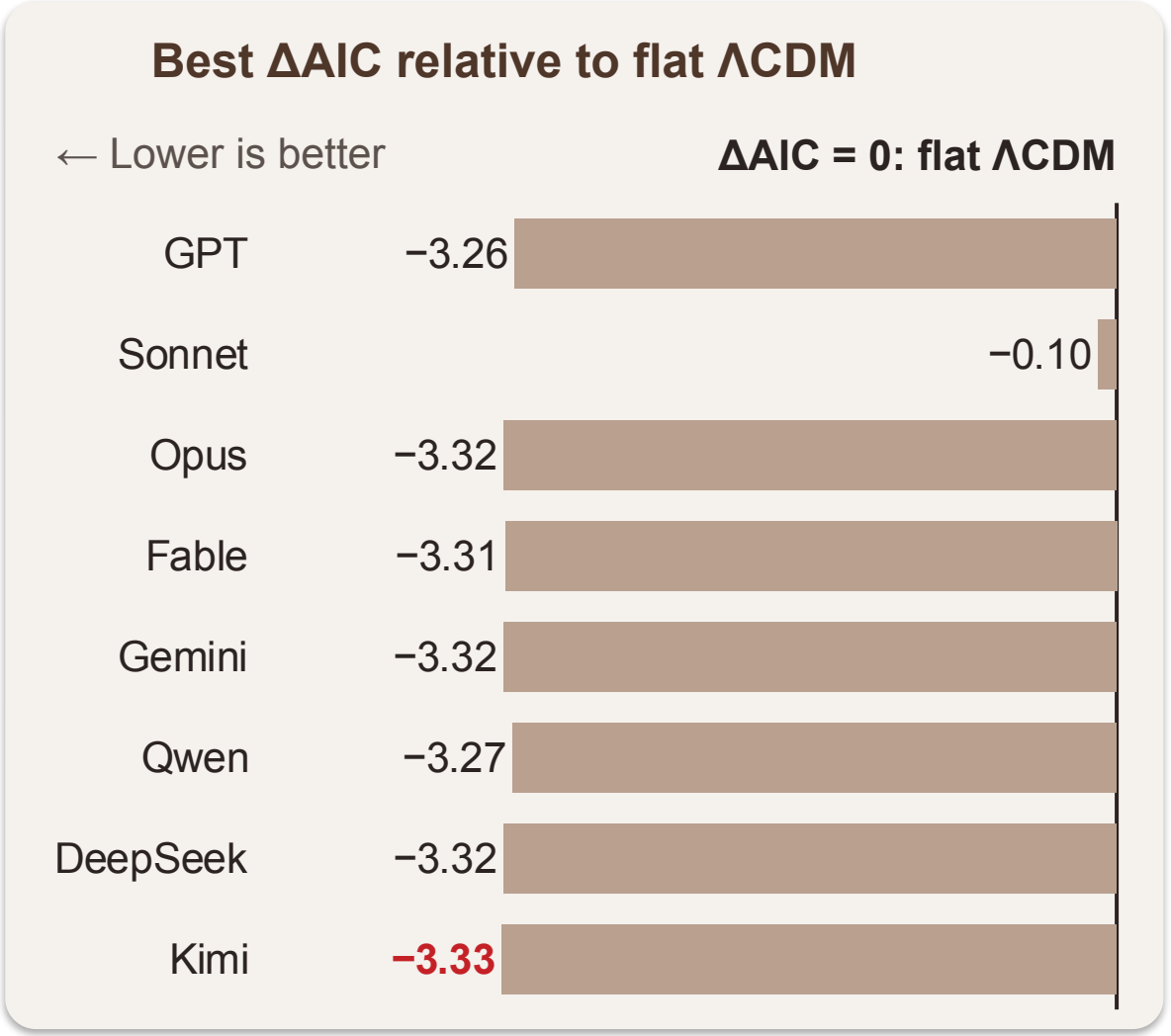
LLM weights remain fixed.

$$p_t(\mathscr{L}, \theta) = \pi_t(\mathscr{L})\, q_t(\theta \mid \mathscr{L})$$

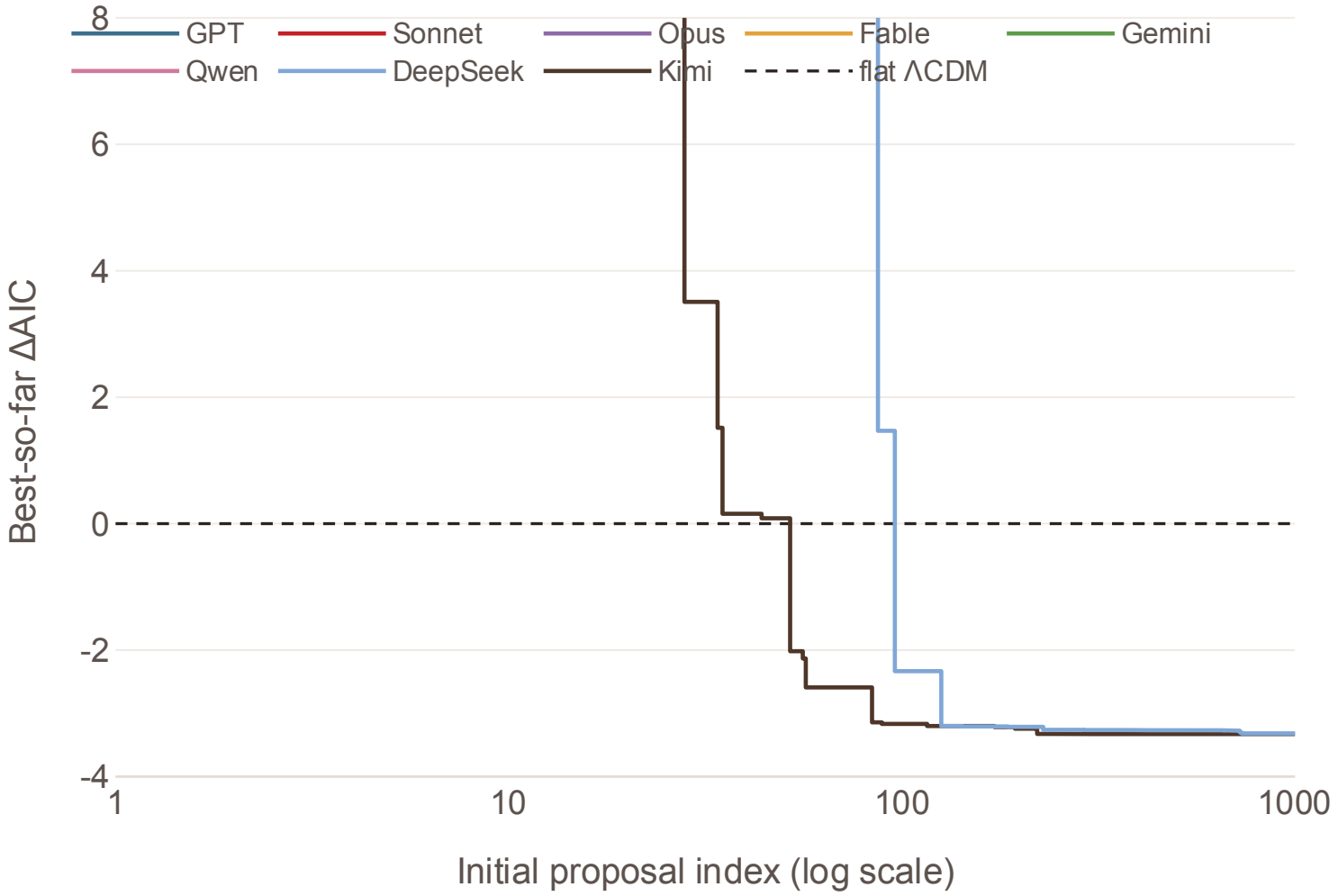


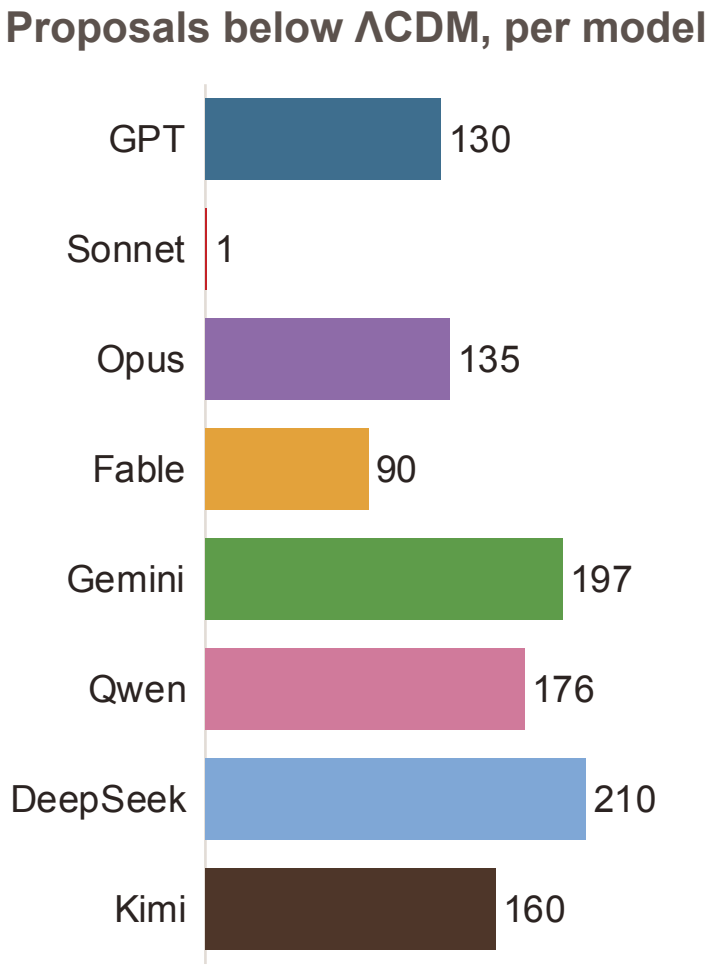
LangGraph manages shared state and conditional transitions.

Result: Eight LLMs, 1,000 proposals each



Searching process

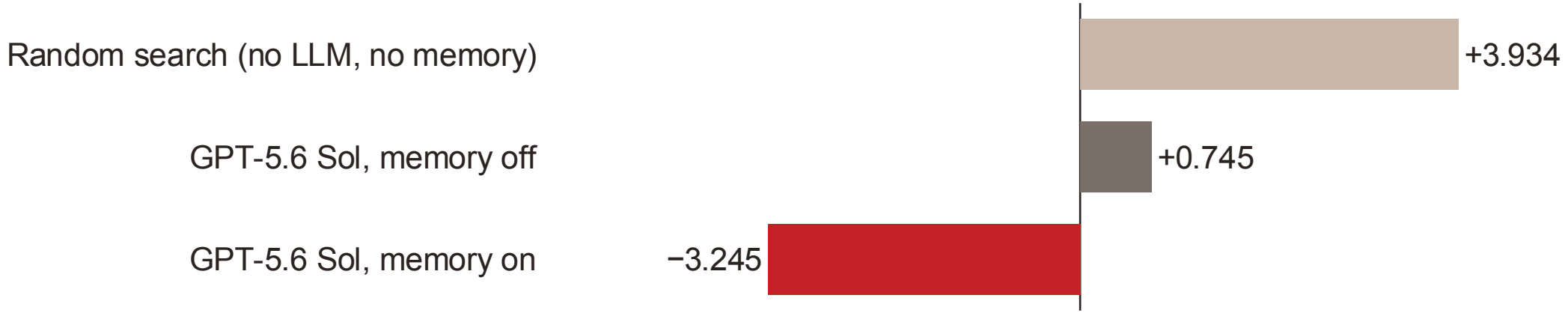




Each initial proposal is counted once, using its best AIC across all evaluated scans.

Random search and memory ablation

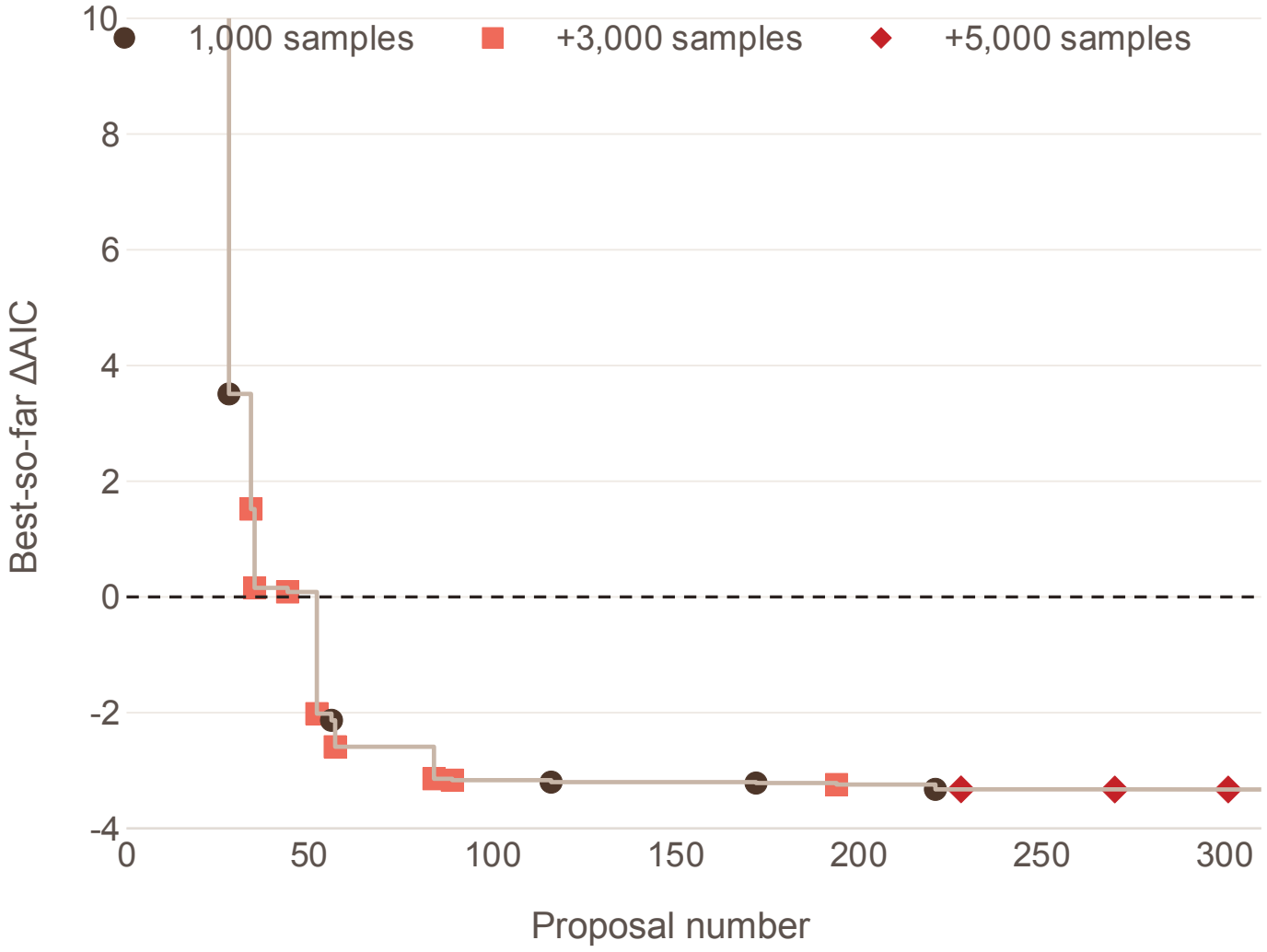
Best ΔAIC relative to flat ΛCDM



Search setting	Proposals	Best ΔAIC	Initial acceptance
Random grammar search (no LLM, no memory)	1,000	+3.934	10.3%
GPT-5.6 Sol, memory off	250	+0.745	39.6%
GPT-5.6 Sol, memory on	250	−3.245	38.4%

Memory-enabled run: 3.99 lower best AIC at similar initial acceptance.

One search in detail: Kimi K3



#3 – #24
Two-field quadratic (ΔAIC ≈ 2000), then cosine (axion) fields: ΔAIC → 24

#26 – #44
One scalar: axion potentials take turns; ΔAIC → 0.1

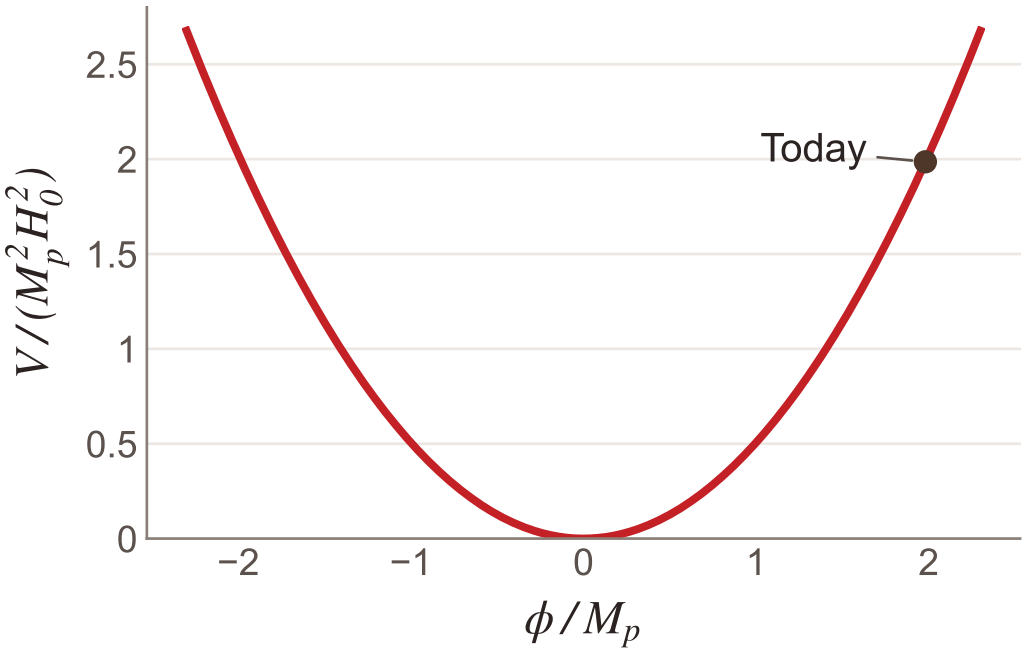
#52 – #57
Mass-term scalar; extra curvature terms. First ΔAIC < 0

#84 – #270
Same model; 3,000- and 5,000-sample scans refine the fit: ΔAIC → −3.33

#301
R + Riemann³ term, set to zero by the filter → quadratic quintessence, −3.328

Recovery of quadratic quintessence

$$\mathcal{L} = \frac{M_p^2}{2} R - \frac{1}{2}(\nabla\phi)^2 - \frac{1}{2} m^2 \phi^2$$



Potential energy drives acceleration

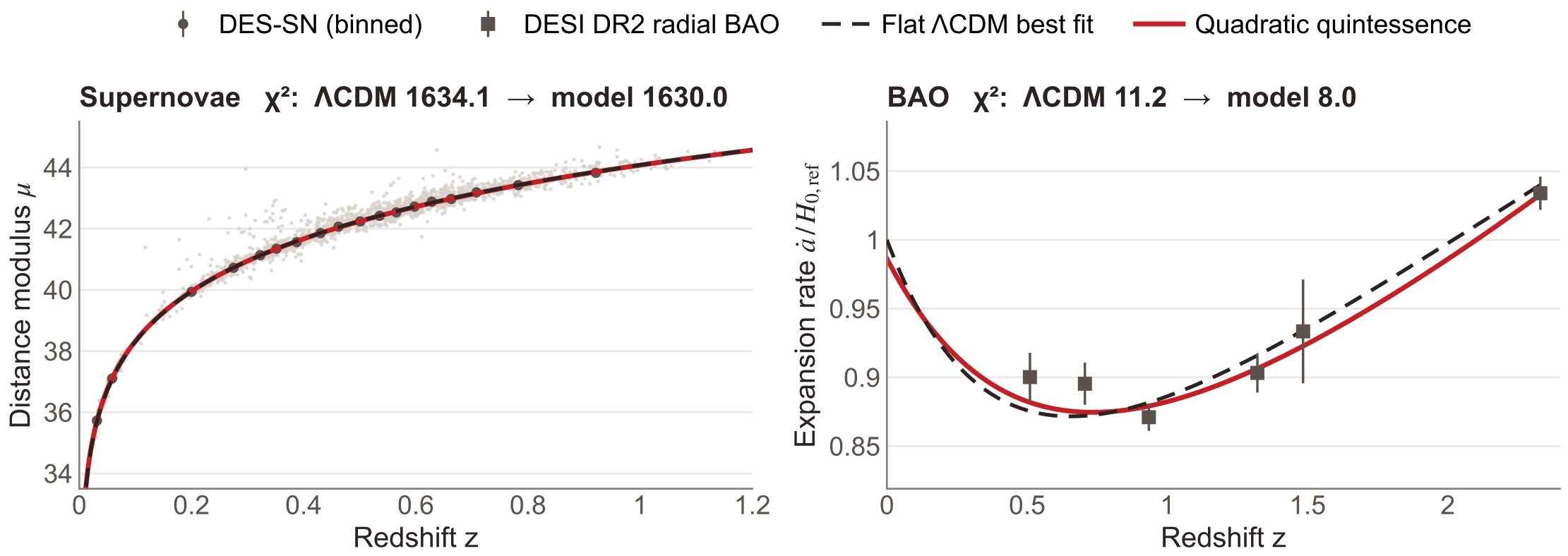
$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V, \qquad p_\phi = \frac{1}{2}\dot{\phi}^2 - V$$

$$\ddot{\phi} + 3H\dot{\phi} + m^2\phi = 0$$

Best fit: $\varphi_0 \approx 2\ M_p$, $m \approx H_0$, $\Omega_\varphi \approx 0.73$
Kinetic / potential ≈ 0.1 today → $w_0 \approx -0.82$

$m / H_0 \approx 1$: the scalar evolves on the cosmological expansion timescale.

Fit improvement and comparison



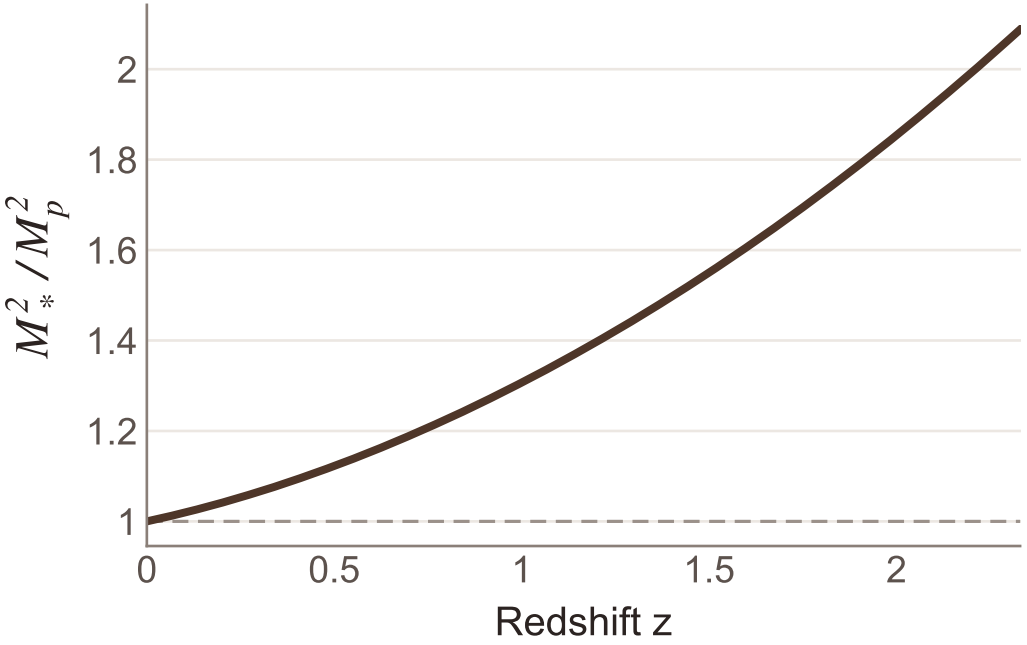
−3.328
ΔAIC relative to flat ΛCDM

−7.328
Δχ²: improvement in the fit

+7.70
ΔBIC: favors ΛCDM

Beyond the EH: induced gravity

$$\mathcal{L} = \frac{1}{2} M_*^2(\phi_2) R - \frac{1}{2} K_{IJ} \nabla_\mu \phi_I \nabla^\mu \phi_J - V$$



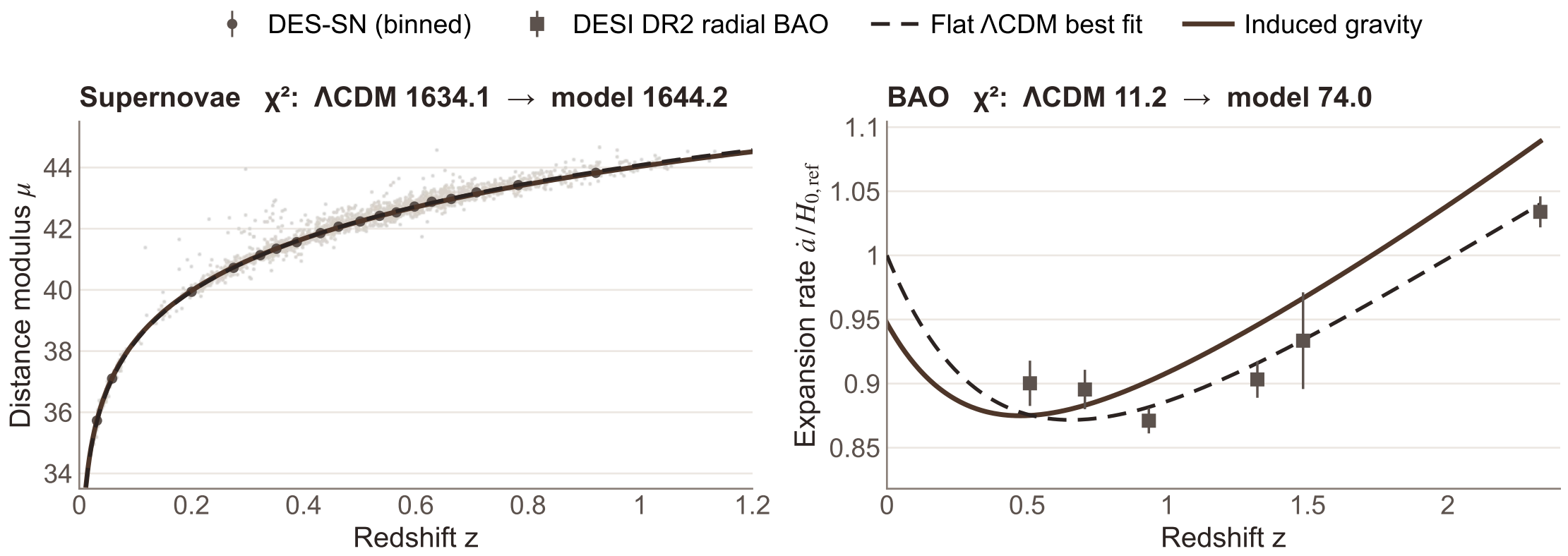
$$M_*^2 = 2\alpha M_p^2 \frac{\phi_2^2}{\Lambda^2}$$

Changing M*² contributes to the Friedmann equation:

$$3M_*^2 H^2 = \rho_m + \frac{1}{2} K_{IJ} \dot{\phi}_I \dot{\phi}_J + V - 3H\dot{M}_*^2$$

A changing effective Planck mass changes the expansion equations.

Induced gravity: poor BAO fit and a scalar ghost



+94.97
ΔAIC relative to flat ΛCDM

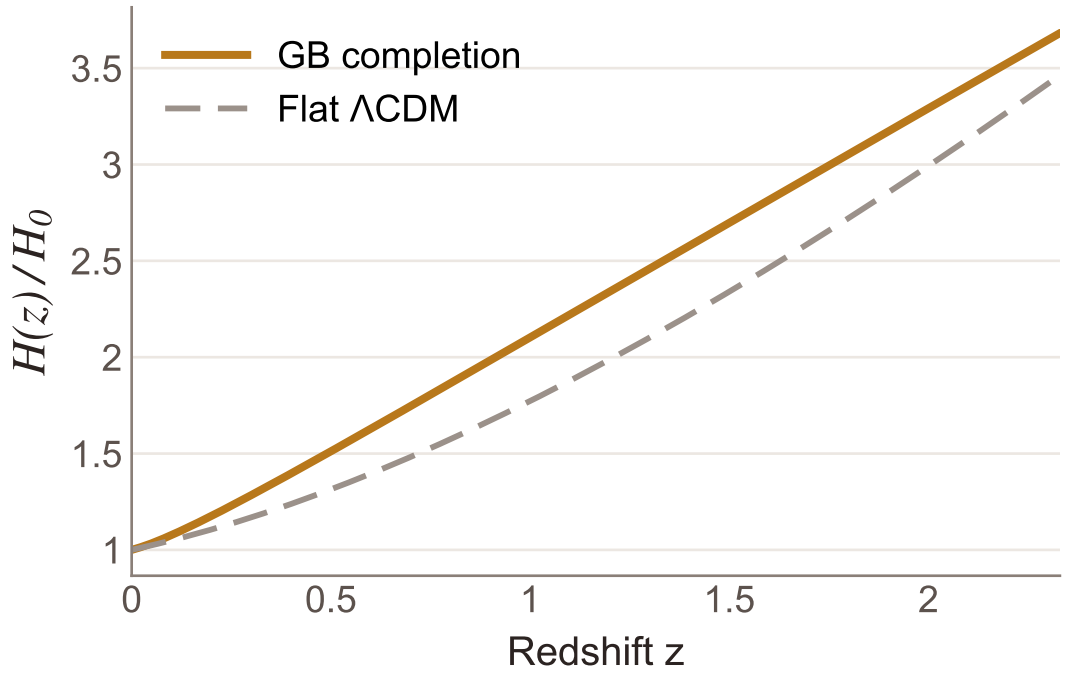
74.05
BAO χ^2 (ΛCDM: 11.2)

Ghost
Negative scalar kinetic term

Beyond the benchmark: A scalar–Gauss–Bonnet theory

$$\mathcal{L} = \frac{1}{2}F(\phi_2)R + \xi(\phi_2)\mathcal{G} - \frac{1}{2}Z(\phi_2)(\nabla\phi_2)^2 - V$$

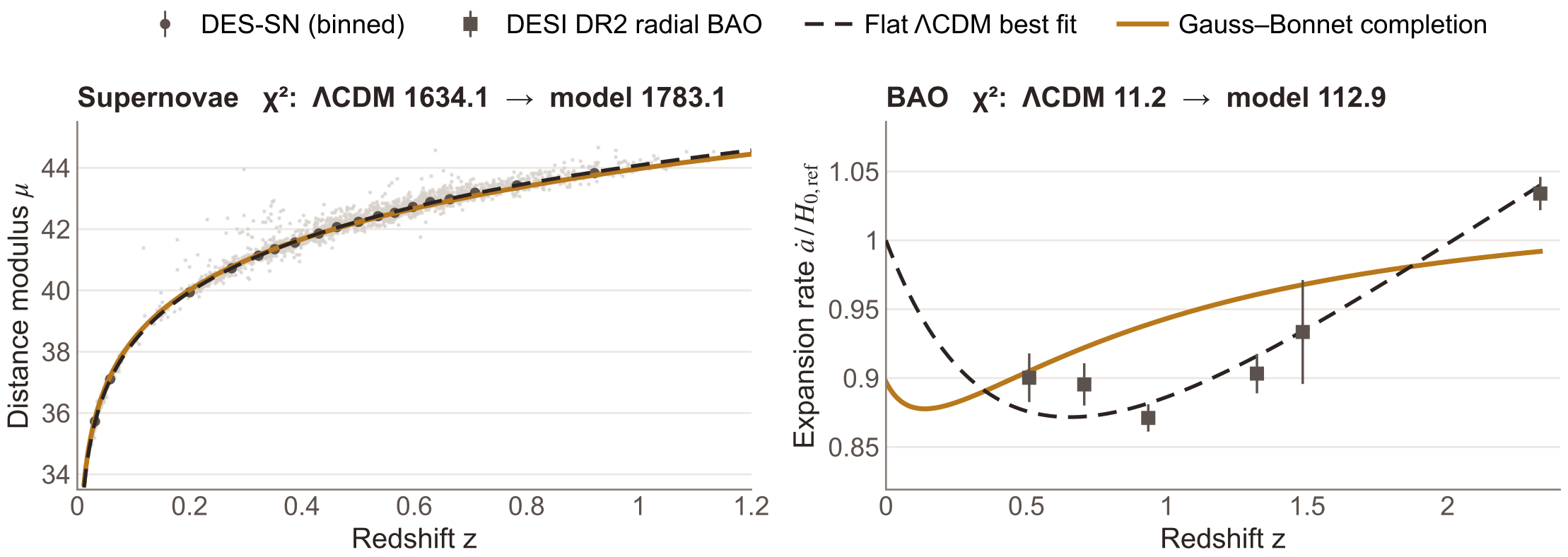
$$\mathcal{G} = R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$$



$$F = 2\gamma M_p^2 \frac{\phi_2^2}{\Lambda^2}, \qquad \xi = -\frac{\varepsilon M_p^2 \phi_2}{2\Lambda^3}$$

$$Z = 1 + 2\eta - 2\theta\frac{\phi_2^2}{\Lambda^2}, \qquad V = \frac{1}{2}m_2^2\phi_2^2$$

Beyond the benchmark: A scalar–Gauss–Bonnet theory



1896.0
Total χ^2

11
Nominal fitted parameters

+266.7
ΔAIC relative to flat ΛCDM

CosmoAgent: results and next steps

1

An automated action-to-data search

Covariant candidates, symbolic equations and observed distances

2

Quadratic quintessence emerges in seven searches

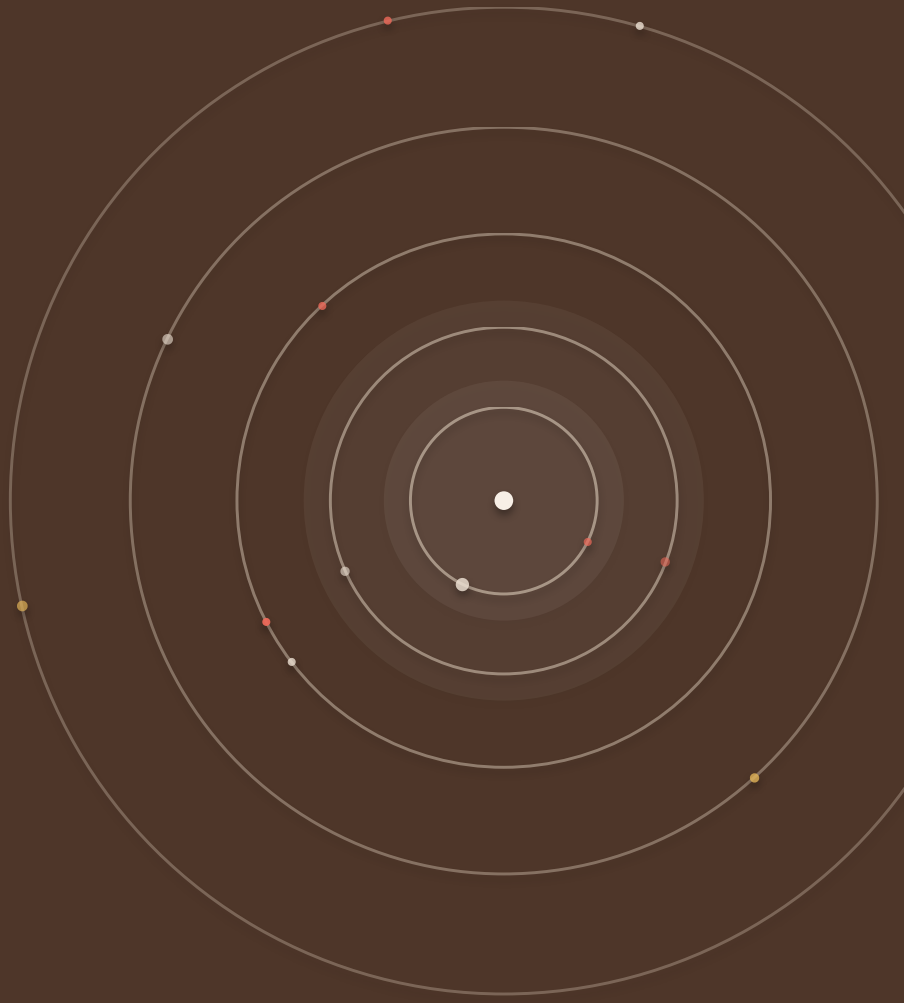
A modest AIC improvement over flat ΛCDM

3

Physical diagnostics guide the next stage

Perturbations and local gravity complement the background fit

Thank you



github.com/wsunamoutlook-prog/AI-Modified-Gravity

arXiv:2610.16536 · wenrong_sun@brown.edu